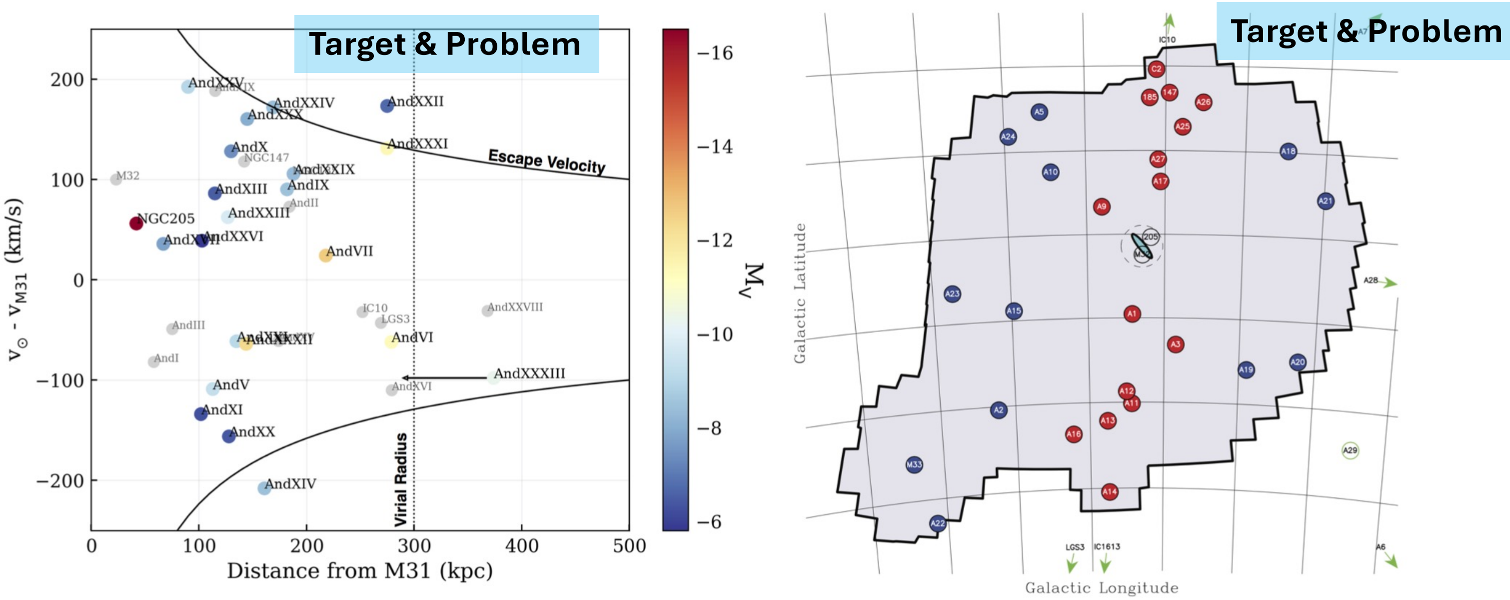
# Figure Captions &
# Talk Abstracts

# Recall: Weisz, goal statement (Giraffe version)

*Through this Treasury program, we propose a comprehensive ACS/WFC3 survey of the M31 satellite system, enabling unique and transformative HST science including critical issues in galaxy formation, cosmic reionization, and the nature of dark matter. Our imaging will target all known satellites within ∼300 kpc of M31, encompassing its full halo profile out to the virial radius. It will have the depth to measure robust SFHs, the cadence to provide secure and precise distances using RR Lyrae, and the astrometric precision for optimal first-epoch proper-motion measurements, all comparable in quality to current measurements of the entire MW system.*

# Weisz




Target

FIGURE 1: *M31 dwarf galaxies in context:* ***Left–*** *Distances and LOS velocities of M31 satellites relative to M31. Our proposed sample is color-coded by luminosity. Current distances are generally highly uncertain (e.g., Conn et al. 2012), highlighted by uncertainty in And XXXIII's location inside/outside M31's virial radius. Virtually all of these satellites are gas-poor, underlining the importance of the M31 halo environment in dwarf galaxy evolution.* ***Right–*** *The spatial distribution of all known M31 satellites as of 2013 (Ibata et al. 2013), with putative members of the thin plane in red and nominal non-plane members in blue. We will observe 23 systems that do not have existing deep HST imaging. Full phase-space information, detailed SFHs, and RR Lyrae-based distances provided by our observations will allow us to investigate the membership, coherence, and origin of the plane, as well as rewind the clock on each galaxy and study the effects of reionization and environment on its evolution.*

Goal

Broader Impact

Fan: JWST, Large, Cycle 3. COSMOS-3D , successful

Recall: goal statement, Javelina version

*A large legacy wide-field spectroscopic survey* is, thus, what is critically needed. **We propose to perform a slitless spectroscopic survey 'COSMOS-3D' with NIR-Cam/WFSS targeting the COSMOS-Web field together with deep parallel MIRI imaging**. This will establish a treasury dataset covering the largest ever field (0.33 $\mathrm{deg}^2$) in the infrared, with secure grism redshifts, enabling countless scientific investigations by the community for the next decade. The survey uses WFSS F444W ($\sim$3.9–5.0 $\mu$m) to cover

# Fan: JWST, Large, Cycle 3. COSMOS-3D , successful

Strategy

Importance


Target

| Total survey area | Total survey volume | # of spec-z galaxies at 7<z<9 | # of spec-z galaxies at 5<z<7 | # of spec-z galaxies at z<5 | # of spec-z AGNs at z>5 | # of SED AGNs at z>5 | # of AGNs at 0<z<5 |
|---|---|---|---|---|---|---|---|
| ~1200 arcmin$^2$ | ~3×10$^7$ Mpc$^3$ | ~800 | ~3600 | ~15000 | 50-100 | 300-500 | ~4000-6000 |

Broader Impact/Take away Statement

Importance

Goal

Figure 1: **COSMOS-3D will enable a wide range of investigations in extragalactic astrophysics and cosmology.** We will address the three most critical science questions in the area of galaxy formation & large-scale structures, supermassive black holes, and cosmic reionization. This legacy program will enable the community to study galaxy assembly from $z \sim 0$ to $z \sim 9$ in unprecedented detail by providing an extraordinary large sample of galaxies and AGNs with spectral measurements, multi-band photometry and images.

PI: Jinyi Yang
Xiaohui Fan

# Monster in the Early Universe: Unveiling the Nature of a Dust Reddened Quasar Hosting a Ten-Billion Solar Mass Black Hole at z=7.1

**Statement of what the figure shows**

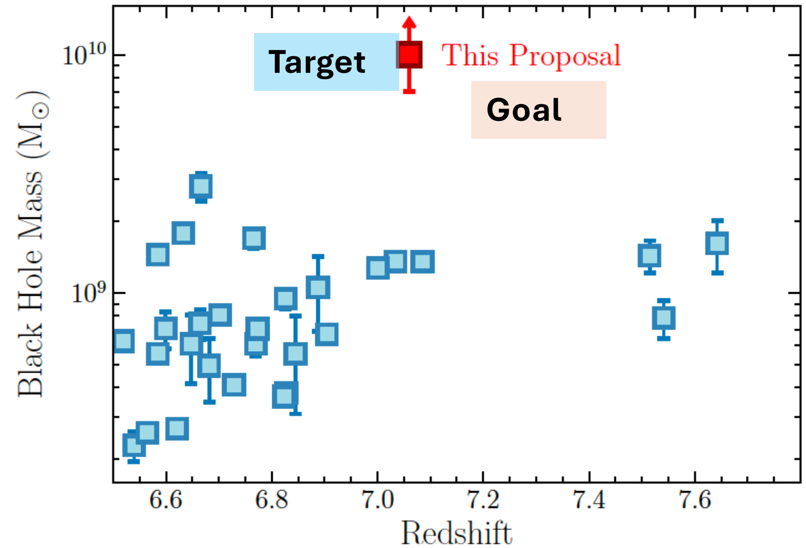
**Target**

**Problem**

**Goal**

**Broader Impact & Importance of solution**

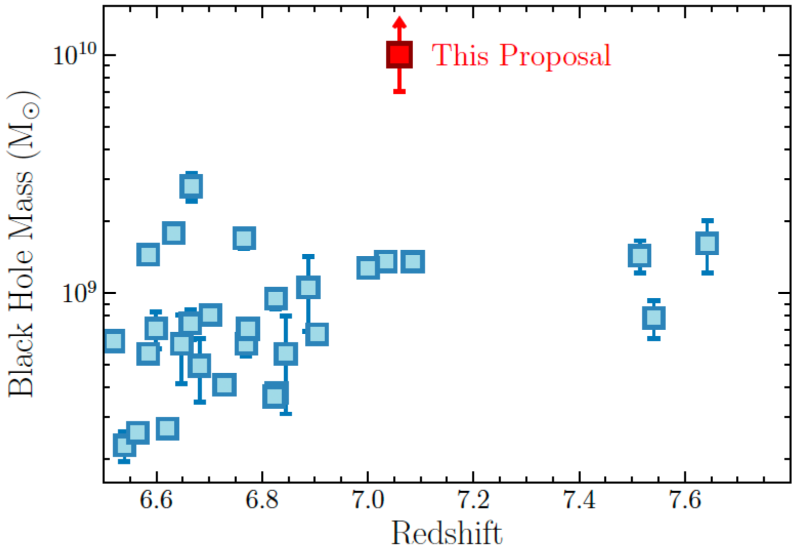
Figure 2: The BH mass vs. redshift distribution of the new quasar J0038–0653 and other known quasars at $z > 6.5$. The current data provide a lower limit of the BH mass of J0038–0653, which is already the most massive at $z > 6.5$. It thus requires a more massive seed BH than any other known quasars under the same assumptions of BH growth. Confirmation of its BH mass will test the limit of early SMBH growth and place the most tight constraint on SMBH formation.



# How could the caption be improved?

PI: Jinyi Yang
Xiaohui Fan

Figure 2: The BH mass vs. redshift distribution of the new quasar J0038–0653 and other known quasars at $z > 6.5$. The current data provide a lower limit of the BH mass of J0038–0653, which is already the most massive at $z > 6.5$. It thus requires a more massive seed BH than any other known quasars under the same assumptions of BH growth. Confirmation of its BH mass will test the limit of early SMBH growth and place the most tight constraint on SMBH formation.



# How could the caption be improved?

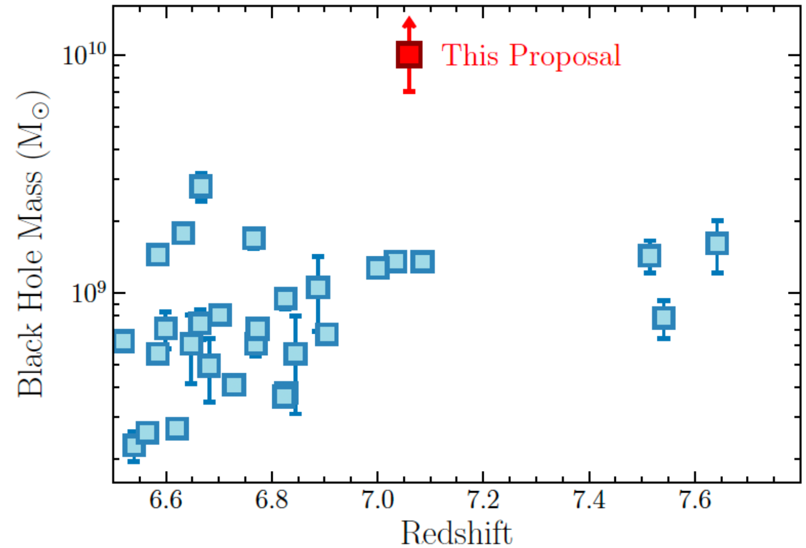
ADD :
Opening Statement, in boldface

**Broader Impact of program**

**“New constraints on early SMBH growth and formation.”**

PI: Jinyi Yang
Xiaohui Fan

Figure 2: The BH mass vs. redshift distribution of the new quasar J0038–0653 and other known quasars at $z > 6.5$. The current data provide a lower limit of the BH mass of J0038–0653, which is already the most massive at $z > 6.5$. It thus requires a more massive seed BH than any other known quasars under the same assumptions of BH growth. Confirmation of its BH mass will test the limit of early SMBH growth and place the most tight constraint on SMBH formation.




**Be clearer about the Goal statement:**

This program will confirm the BH mass

Example from Xiangyu Jin's proposal from 2022 for this class.

Note: Figure caption missing the broader impact– but this is an example of a relevant figure format.

**Problem**

**Goal Statement**

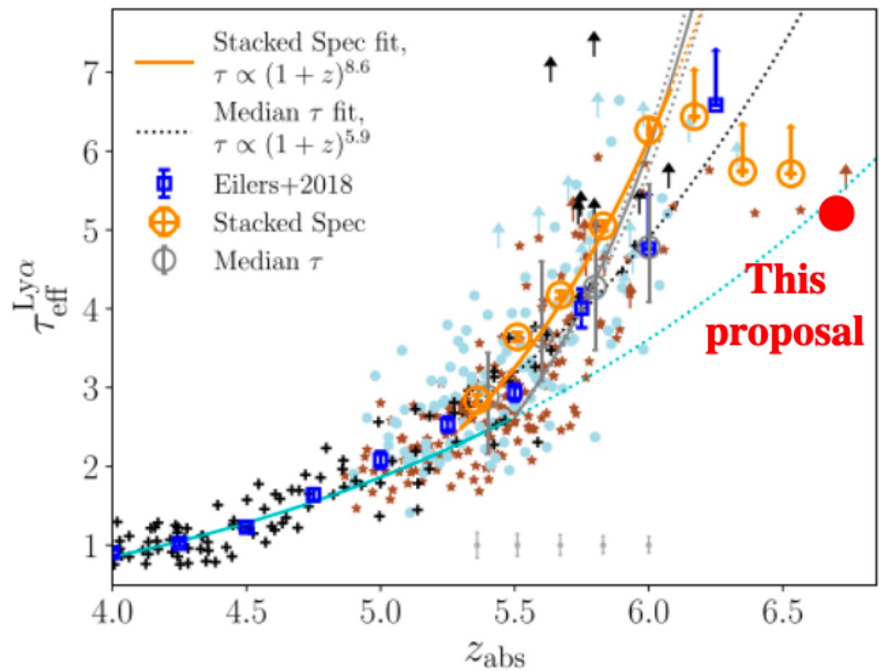



Figure 3: Measurements of Ly$\alpha$ effective optical depth (points) and fitted Ly$\alpha$ effective optical depth redshift evolutions (lines) from Yang et al. (2020a). We will verify the nature of the $z = 6.7$ transmission spike and measure its optical depth in the Ly$\alpha$ forest with the proposed WFC3/UVIS imaging. The expected Ly$\alpha$ optical depth obtained by this program is denoted by the red circle, assuming only the $z = 6.7$ transmission spike contributes to the flux inside the FQ937N filter and the flux density is zero elsewhere. The expected Ly$\alpha$ effective optical depth of the $z = 6.7$ transmission spike is much lower than the predicted value from Yang et al. (2020a) (the orange line). With the proposed program, we will be able to verify the nature of the transmission spike and to confirm the timing of one of the first highly ionized patches appearing in the IGM.

Example from Claire Cook's proposal from 2022 for this class.

Note: Figure caption is missing the bigger picture – but this is an example of a relevant figure format.

**Problem**

**Goal Statement**

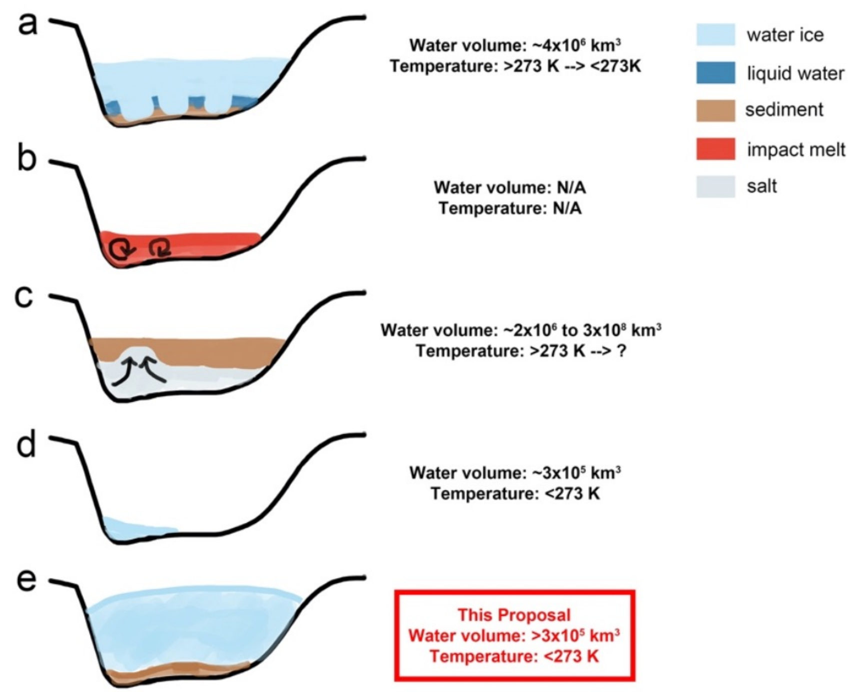



Figure 1: Cross-sections of Hellas basin (not to scale), showing proposed banded terrain formation scenarios and implications for water volume and surface temperature required. These hypotheses remain debated because observations alone allow for multiple interpretations. a) Deformation of sediment by ice blocks in an ice-covered lake (Moore & Wilhelms 2001). b) Convection in impact melt (Kite + 2009). c) Sediment layers deformed by upwelling salt emplaced by large volumes of saline water (Mangold & Allemand 2003). d) Deformation of a thin ice-rich layer (Diot + 2015). e) Deformation of sediment beneath an ice sheet (Bernhardt + 2019). Scenario (e) is the leading hypothesis, in which the ice overburden, in conjunction with varying basal topography and basal properties may lead to a complex stress field consistent with the complex pattern of deformation observed in the banded terrain. **This proposal will test this leading hypothesis, and thus contribute to resolving the problem of the banded terrain's origin.**

# Assign 4 Part A.  Figure Caption format

- First sentence: Bold faced takeaway that highlights the broader impact of the program.
- Second sentence: Description of what is plotted
- Next:  Statement of the problem related to the target/plot
- Last sentence(s) is(are) the Goal  ("This program will …" or "We propose to") AND the Importance of the solution.  Underline the Goal statement.

# Talk Abstracts & Elevator Pitch

Assign 4 Part B → turn the figure caption into a talk abstract

# Goals of a talk abstract?

- Convince someone to come to your talk!
- Come talk to you more/look at your paper in more detail – even if you didn't attend the talk.
- Conference → Convince them to give you a talk.
- Job talk → Convince them to give you a job
- Balance the amount of detail/length- want to leave something for the talk.

# Goals of a talk abstract?

- <u>Get folks to come to your talk</u>
- Convey an exciting problem / exciting new aspect of your subfield
- Convey that you will help folks learn something new
- Convey that there are future directions for your subfield that are exciting

- Be concise

# Goals of an elevator pitch?

- Your name is jargon at the beginning- you have to add context to who you are and what you want to do.

# Goals of an elevator pitch?

- Get folks to be **excited about your research**
- Convey an exciting problem / exciting new aspect of your subfield
- Convey that you will help folks learn something new
- Convey that there are future directions for your subfield that are exciting

- Be concise

**“folks” here is the field/the public/the person you are talking to.**
**You need to use “I”, “my research”**

**Job talk abstracts are elevator pitches.**
**All talks are job talks.**

**LPL COLLOQUIUM** Tuesday, November 9 2021
**Dr. Tyler D. Robinson**
Northern Arizona University

<u>Towards New Worlds: Exploring Exoplanets with Next-Generation Missions</u>

Thousands of exoplanets have been discovered to date, many with sizes or masses unencountered in our Solar System. Excitingly, we are entering into an era where a diversity of observational resources will be able to tell us what these distant worlds are like. Amongst such endeavors, exoplanet direct imaging presents a critical and well-understood path towards exploring worlds that may even be Earth-like. The development of direct imaging mission concepts is an exciting step forward for the field of exoplanet science, and this presentation will cover the science that would come from such a mission in the areas of planetary studies, habitability, and life detection.

**Fact**

**Problem/New Discovery (title)**

**Support/Importance of Problem/New Discovery (specific to talk)**

**Future Outlook**

**Overview of Talk**

Talk doesn't have to include a "Problem" per say

**Dr. Tyler D. Robinson**

How would you change this into an elevator pitch?

**GOAL TO CONVEY**: Towards New Worlds: Exploring Exoplanets with Next-Generation Missions

Thousands of exoplanets have been discovered to date, many with sizes or masses unencountered in our Solar System. Excitingly, we are entering into an era where a diversity of observational resources will be able to tell us what these distant worlds are like. Amongst such endeavors, exoplanet direct imaging presents a critical and well-understood path towards exploring worlds that may even be Earth-like. The development of direct imaging mission concepts is an exciting step forward for the field of exoplanet science, and this presentation will cover the science that would come from such a mission in the areas of planetary studies, habitability, and life detection.

**Dr. Tyler D. Robinson**

My Elevator Pitch version

**GOAL TO CONVEY**: Towards New Worlds: Exploring Exoplanets with Next-Generation Missions

Thousands of exoplanets have been discovered to date, **many of them have** sizes or masses **that are different from our** Solar System. **What's really exciting is** that **there are now many new** observational resources **that will tell** us what these distant worlds are like. **My research focuses on** exoplanet direct **imaging as a method to explore** worlds that may even be Earth-like. **By leading the** development of direct imaging mission concepts, **my research will advance our understanding of habitability and how we might detect life on other worlds.**

**LPL Colloquium** Thursday, December 2, 2021
**Dr. Sukrit Ranjan**
Northwestern University

**<u>Theoretical Underpinnings of the Search for Life on Exoplanets</u>**

We are on the cusp of a revolution in our understanding of life in the universe. Exoplanet surveys like *Kepler* have revealed potentially habitable planets to be common, and upcoming facilities like the *James Webb Space Telescope* and the *Thirty Meter Telescope* will at last have the ability to characterize their atmospheres in search of signs of life. However, proceeding from the astrophysical observables to inferences regarding the presence or absence of life will require considerable theoretical intervention. In this talk, I will illustrate the theoretical infrastructure that must be developed to prepare for exoplanet life search with three vignettes. Specifically, I will discuss (1) the UV environment on planets orbiting M-dwarfs and the implications for their inhabitability, (2) the photochemistry of water vapor and the implications for oxygen as a biosignature gas on planets orbiting Sun-like stars, and (3) efforts to increase the catalog of potential remote biomarkers of life, with a focus on reactive gases like phosphine. I will conclude by reviewing the considerable theoretical and laboratory work that remains to be done to prepare for the era of habitable planet characterization.

**Fact & Imp** | **Problem/New Discovery (title)** | **Overview of Talk (strategy)** | **Future Outlook**

**Steward Colloquium: Tuesday Feb 28th 2023**

**Dr. Carl Fields**
Los Alamos National Laboratory

Title: The Next Generation of Stellar Astrophysics

**Abstract:** Stars play a critical role throughout our Universe via galactic chemical evolution, compact object formation, and stellar feedback. Computational models of stars have progressed over the decades in concert with astrophysical observations to advance our understanding of stellar evolution and transient phenomena. These models have benefitted from new observational campaigns, nuclear physics experiments, and computational and technological advancements. However, despite these advancements, there are long-standing challenges among stellar models that require novel approaches. I will discuss my work leveraging one-dimensional (1D) stellar models, 3D stellar convection simulations, and models of stellar transients to move us towards the next generation of stellar astrophysics.

Facts

Problem/New Discovery (title)

Overview of Talk

Future Outlook

**Dr. Carl Fields**

**Full job talk advert – also good example for a conference application where more details needed**

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